Movie2Comics: A Feast of Multimedia Artwork

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ABSTRACT

As a type of artwork, *comics* are prevalent and popular around the world. However, although there are several assistive software and tools available, the creation of comics is still a tedious and labor intensive process. This paper proposes a scheme that is able to automatically turn a movie to comics with two principles: (1) optimizing the information reservation of movie; and (2) generating outputs following the rules and styles of comics. The scheme mainly contains three components: script-face mapping, key-scene extraction, and cartoonization. Script-face mapping utilizes face recognition and tracking techniques to accomplish the mapping between character's faces and their scripts. Key-scene extraction then combines the frames derived from subshots and the extracted index frames based on subtitle to select a sequence of frames for cartoonization. Finally, the cartoonization is accomplished via four steps: panel scale, stylization, word balloon placement and comics layout. Experiments conducted on a set of movie clips have demonstrated the usefulness and effectiveness of the scheme.

Categories and Subject Descriptors

H.5.1 [**Information Interfaces and Presentation**]: Multimedia Information Systems-Evaluation/methodology; C.4 [**Performance of Systems**]: Design studies;

General Terms

Experimentation, Performance

Keywords

Comics, Key-scene Extraction, Cartoonization

1. INTRODUCTION

Comics are a graphic medium where pictures convey a sequential narrative while speech is in the form of text in balloons. Today, comics can be found in newspapers, magazines, graphic novels, even on the web and its conventions

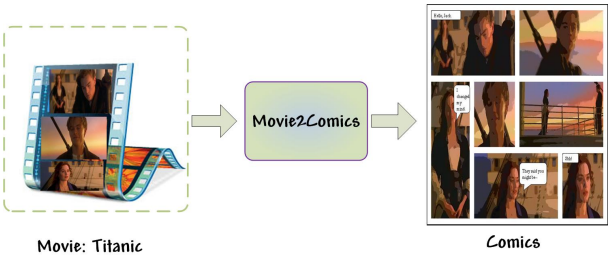


Figure 1: Schematic illustration of *movie2comics*.

were developed differently around the globe. However, the creation of comics seems inefficient with respect to its prevalence and popularity. One of the principal factors is because the comic artists are accustomed to manually composing the artwork using traditional tools such as pencil, ink, brush, etc. Another factor is that the rapid development of computation technology merely transfer the creation from paper and pencil to computer and mouse, *i.e.*, computer aided design but not automatical composition. Therefore, intense human labor is still involved in the the process of comics' creation.

Is it possible to automatically generate the artwork of comics? As we know, producing picture and text in comics that communicate the right message is the artist's job. Artificial intelligence is still far from an artist's capability and expressiveness [4][15]. As a tradeoff, recently there emerge two works that tackle the problem by turning movie to comics. The first one is cartoon generation from video stream [5]. It manually selects frames with more important feature and transforms them into simplified illustrations. Stylized comic effects including speed line, rotational trajectory and the background effects, are inserted into each illustration while word balloons are automatically placed. Further work in [2] seeks an automated approach of word balloon placement based on a more in-depth analysis of comic grammars. The second work [10] employs the screenplay of movie for turning movie into a comic strip since the screenplay information is able to offer important clues to segment the film into scenes and to create different types of word balloons. However, these two works are semi-automatic and can still be categorized into "computer aided design", such as the manual selection of important frames in [5] and word balloon placement and comic layout re-arrangement in [10]. Furthermore, a significant issue is not touched in their methods: how to identify who is the speaker? Especially in the situation of when multiple characters are involved in a single frame or the speaker is occluded. Therefore, the main challenges in turning movie into comics are twofold. The first is

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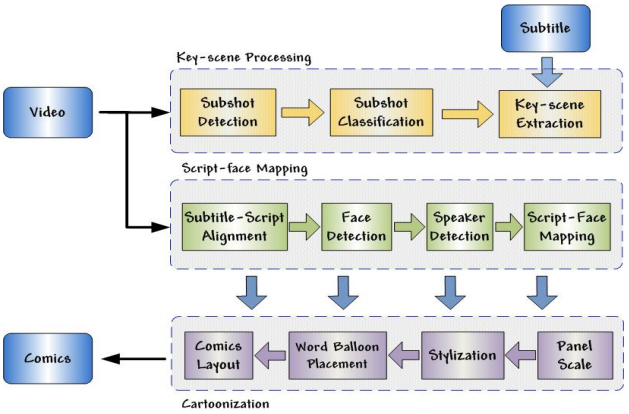


Figure 2: The system framework of the *movie2comics*

to recognize the speaker and the second is to automatically cartoonize the sequence of the extracted "key-scenes"[1].

In this paper, we propose a scheme which can automatically turn movie to comics, namely *movie2comics*(see Fig. 1). The main contributions of this paper are: 1) to the best of our knowledge, this is the first work towards realizing automatic conversion of movie into comics; 2) we design an automatic script-face mapping algorithm to identify the speaker in the key-scenes where multiple characters are involved; 3) we propose a viable method for panel (refer to as a single picture within comics, i.e., the extracted key-scenes in this scenario) scale and organize panels according to the traditional layout of comics.

2. SYSTEM FRAMEWORK

Since that two principles should be considered in the system design of *movie2comics*: retain as much informative content within movie as possible and stylize the extracted key-scenes according to the rules in the creation of comics, we propose the framework of *movie2comics* as illustrated in Figure 2. There are three main components for realizing different functionalities according to the above two principles. Script-face mapping is designed to attach the speech content around the human faces, which agrees with the specific expression in comics. In key-scene extraction, subshot classification is selected as the basic unit bacause shot is usually too long and contains diverse contents. The extraction utilizes the index frames extracted based on aligned subtitle and the information from classified subshots. After that, a series of cartoonization processes are carried out on the extracted key-scenes.

3. SCRIPT-FACE MAPPING

In this section, script-face mapping is presented to recognize speakers in the movie and map them to the scripts. We utilize the method in [3] to merge the speech content, speaker identity and time information from subtitle and script and then implement a face detector [13] to extract faces from the frames in the speaking parts. After that, lip motion analysis [16] is employed to tackle the script-face matching to establish whether the character is speaking when the frame only contains one face based on the fact that

[1]In this study, we use the term *"key-scene"* indicates the retained informative **frame** but not the higher level structure of shot in video structure

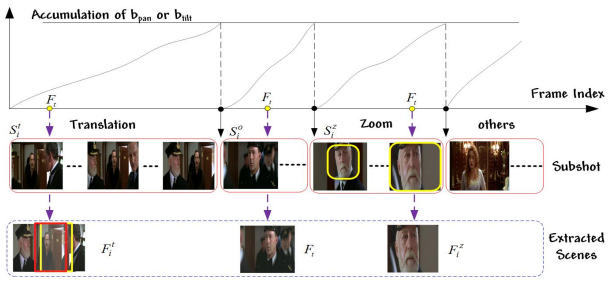


Figure 3: An example of key-scene extraction from the movie "*Titanic*". It includes the subshots of *translation*, *zoom* and others as well as the process of generating mosaic images.

speaking is associated with distinctive lip movement. In the cases that a frame contains more than one face. Our approach is to first label faces with speaker identities and then match them with scripts accordingly (the script file contains the speaker identity information). The highly-confident labeled tracks are treated as training exemplars to predict other tracks that are unlabeled due to not containing enough established identities. Each unlabeled face track is simply represented as a set of history image feature vectors.

In this work, by regarding the identifying of each history image in a testing face track as a task, we formulate the face track identification challenge as a multi-task face recognition problem. This motivates us to apply the multi-task joint sparse representation model [9] to accomplish the task. We construct the representation of face appearance by a part-based descriptor extracted around local facial features [3]. Here we first use a generative model [1] to locate nine facial key-points in the detected face region, including the left and right corners of two eyes, the two nostrils and the trip of the nose and the left and right corners of the mouth. We then extract the 128-dim SIFT descriptor from each key-point and concatenate them to form a 1152-dimensional face descriptor (SiftFD). After labeling each face track with speaker identity, we can establish the speaking character even there are more than one face in a frame. It is worth mentioning that there also exist scripts that cannot be successfully mapped to faces, and in this work we directly put them on the top of panels (off-screen voice is also processed in the same way).

4. KEY-SCENE EXTRACTION

Key-scene extraction is to extract informative frames, *i.e.*, previously defined key-scenes. The basic idea is to decompose the movie into a series of subshots by a motion based method [8], where each subshot is further classified into predefined categories on the basis of camera motions. An appropriate number of frames or systhesized mosaic images are extracted from each subshot.

Zoom subshot. The subshots can be categorized into *zoom-in* and *zoom-out* based on the tracking direction and b_{zoom} which indicates the magnitude and direction of zoom. In the *zoom-in* subshots (as shown in Figure 3), the first frame is sufficient to represent the whole content. While *zoom-out* is the reverse and the last frame can be the representative. If there is an index frame (the frames marked by the time index in subtitle) falls in this category of subshot, both index frame and representative frame are taken as the key-scenes.

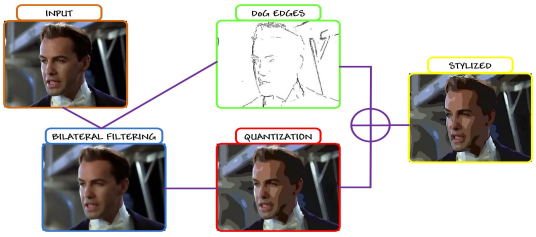


Figure 4: Panel stylization

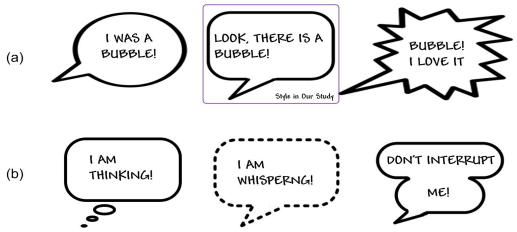


Figure 5: Balloon types

Translation subshot. It represents a scene through which camera is tracking horizontally or vertically. In this scenario, image mosaic is usually employed to describe the wide field-of-view of the subshot in a compact form. Before generating the panoramas for each subshot, we first segment the subshot into units to ensure homogeneous motion and content in each unit [12]. As the wide view panorama is derived from a large number of successive frames probably resulting in distortion in the generated mosaic, each subshot is segmented into units using leaky bucket algorithm [7]. As shown in Figure 3, if the accumulation value exceed T_p/t , one unit is segmented from the subshot and we generate a mosaic image to represent this unit [6]. In this case, we take the mosaic image as the representative even if there appear index frames in these subshots.

5. CARTOONIZATION

In this section, we describe the cascaded steps toward cartoonize the extracted key-scenes in this section where four steps are included: panel scale, panel stylization, word balloon placement and organize panels in a comic-like way.

5.1 Panel Scale

Panel refers to an individual frame in the multi-panel sequence of comics and it consists of one drawing that depicts a single moment. Consider that the size and number of recognized faces in each frame have been recorded, we can scale panels by performing segmentation based on the number and size of recognized faces. Four classes are pre-defined as: cannot segment; segmentation around the face; vertically segment the identified speaker and horizontally segment the roles. We then define their rules according to the ratio of width and height of faces as well as the distances between faces, respectively. It is worth mentioning that if a given frame satisfies the rule, whether to perform segmentation is still constrained by the layout of the comics.

5.2 Panel Stylization

For panel stylization, there are different methods according to generes. We employ the stylization analogous to [14], which is able to abstract imagery by modifying the contrast of visually important features, *i.e.*, luminance and color opponency. The basic workflow of our stylization scheme is shown in Figure 4. We first exaggerate the given contrast in an image using nonlinear diffusion and then add highlighting edges to increase local contrast. We finally stylize and sharpen the resulting images.

5.3 Word Balloon Placement

Word balloon are one of the most distinctive and readily recognizable elements of the comic medium. Its appearance varies dramatically from artist to artist as illustrated in Figure 5(a), which shows three most common types of word balloons. For a given artistic style such as the middle type of Figure 5(a), there are another three types of balloon within the comic vocabulary as illustrated in Figure 5(b) (the style we employed in our system). In our system, all balloons are placed either to the right of character's face, above the character's head, or left of the character's faces. For simplicity and efficiency, the balloons are not generated by graphics technique but only the image layer mask, *i.e.*, we manually create various types of balloon mask as illustrated in Figure 5(b) and the middle of Figure 5(a). These types are able to be scaled, rotated and flipped to meet the requirements in different situations.

5.4 Comics Layout

The initial layout template is designed as illustrated in Figure 6(a) where each row has two panels while the whole page contains three rows. The width and height of the page as well as the intervals between each panel are fixed. There are eight manually pre-defined templates in total, two of which are illustrated in Figure 6(a) and (c). We have eight pre-defined templates which can be deemed as eight sequences according to the reading order, *i.e.*, from left to right and from top to bottom. Figure 6(c) illustrates one of the templates. To enhance the layout diversity, we also define a preference rank of the eight templates. Given the extracted key-scene list and the pre-defined eight templates with their ranks, the method read a sub-sequence with the given length in a decreasing order of rank. It then calculates the Hamming distance between each template and the sub-sequence and terminate at when distance equals to zero.

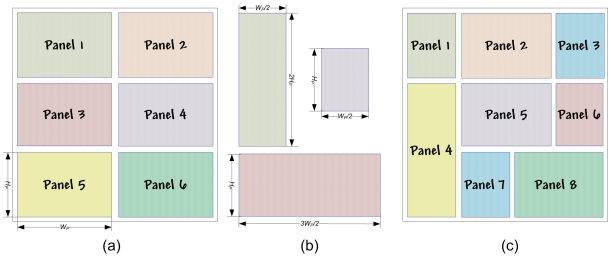


Figure 6: The comics layout. (a) the standard template; (b) panels with another three types of size; (c) one of the eight templates.

6. EVALUATION

We conduct our experiments using 15 movie clips segmented from three movies: *"Titanic"*, *"Sherlock Holmes"* and *"The Message"*. In evaluation, twelve participants (10 males and 2 females with their ages range from 23 to 30) are involved. We convert each movie clip to comics and evaluate them from two aspects: content comprehension and user impression. Content comprehension measures the variation of

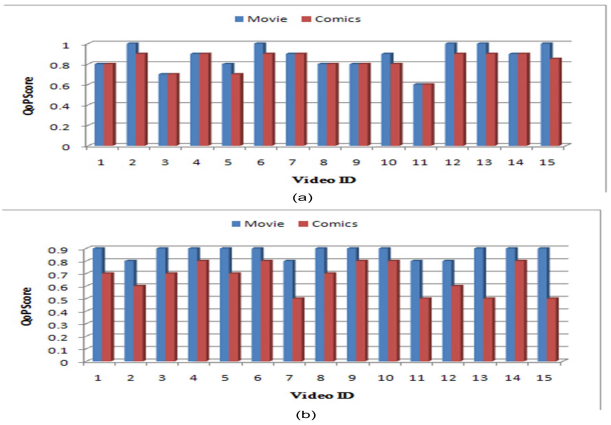


Figure 7: The QoP from (a) the script source and (b) the visual source, respectively.

understanding by users between movie clip and its automatically generated comics while user impression evaluates the user experience in viewing the generated comics based on two criteria: enjoyment and acceptance.

6.1 Content Comprehension

As we know, some questions such as "how many characters are there in this movie clip", *etc.*, have a single definite answer. Thus it is possible to determine what percentage of questions each participant answered correctly. Besides that, some questions may be answered if only certain information is assimilated from specific information sources. For example, the question "who wore the sports clothes numbered 23?", can only be answered by the video text. We can determine the proportion of correctly answered questions which are related to the different types of information. Here, we categorize the sources of the questions into 2 types as follows: **Script:** information from the subtitle only (10 questions in total); **Visual:** information derived from visual content in movie (10 questions in total).

These questions (20 in total) are carefully designed to cover as much details of the content in the movie clips as possible. For performance comparison, we define a metric of *QoP*, Quality of perception as the ratio of the correctly answered questions to total number of the questions. Figure 7(a) and 7(b) illustrate the percentage of correctly answered questions using movie and comics, respectively. The IDs of the movie clips are is identical to the order as presented in Table 2. We can see that the context (i.e., story, which is mainly conveyed by subtitle) is mostly retained after turning into comics. However, the visual information loss is more noticeable (reducing from 87.33 to 66.67). It is argued by comics artists that such "frame of pictures and abstract presentation" enable audience envision richer and extend the content of the story [11].

6.2 User Impression

This section evaluates the user impression in terms of two criteria: *enjoyment* (It measures the extent to which users feel that the comics is enjoyable) and *acceptance* (It gives a score to reflect whether the users like the style). Each user was asked to assign a score of 1 to 10 (higher score indicating better experience) to the above two criteria. Figure 8 shows the averages of the two criteria. We can see that although comics can convey the whole story to audience, however, the

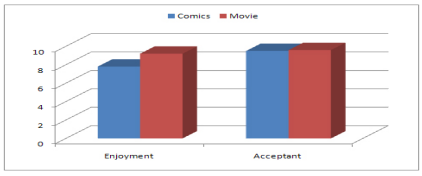


Figure 8: The comparison of user impression.

enjoyment metric is much degraded. It is inevitable due to the loss of multimedia content. In terms of "acceptance", they are close to each other. This indicates that the scheme in this paper is useful.

7. CONCLUSION

We have presented an effective and efficient scheme of turning movies to comics where less than 10s is needed to process a video clip with an average duration of 5 mins on a PC with Pentium 4 3.0G CPU and 2G memory. Furthermore, although the current scheme targets at movies, it can be extended to process TV programs, documentary, etc. Therefore, our proposed scheme can be deemed as a first and effective exploration in this research direction.

8. ACKNOWLEDGMENTS:

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